INCH-POUND

MIL-M-38510/110B
AMENDMENT 3
8 FEBRUARY 1999
SUPERSEDING
MIL-M-38510/110B
AMENDMENT 2
02 JUNE 1998

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, QUAD OPERATIONAL AMPLIFIERS,
MONOLITHIC SILICON

Inactive for new design after 13 July 1995

This amendment forms a part of MIL-M-38510/110B, dated 11 September 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.1, add footnote for device type 01 which will state "Devices may be monolithic or they may consist of two separate independent die."

PAGE 2

* 1.2.3, add the following new case outline:

Outline letter	Case outline (see MIL-M-38510, appendix C)
" Z	See figure 1 "

* 1.5, add the following:

Case outline	Package	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
" Z	14 lead gull wing flat package	350 mW @ T_A = 125°C	25°C/W	176°C/W Still air 116°C/W 500 LFPM "

3/, second line, delete the words inside the parenthesis, and substitute "(36 volts for device types 05 and 06)."

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PAGE 3

3.1, delete and substitute the following;
"3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 4

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Input offset voltage test. Footnote "1/" will be renumbered as footnote "2/".

TABLE I, Input offset current test. Footnote "1/" will be renumbered as footnote "2/".

TABLE I. Input bias current test. Footnote "1/" will be renumbered as footnote "2/". Device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min". Device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

TABLE I, Input voltage common mode rejection test. Footnote "2/" will be renumbered as footnote "3/".

TABLE I. Output short circuit current (for positive output) test. Footnote "3/" will be renumbered as footnote "4/". Footnote "4/" will be renumbered as footnote "5/".

TABLE I. Output short circuit current (for negative output) test. Footnote "3/" will be renumbered as footnote "4/". Footnote "4/" will be renumbered as footnote "5/".

PAGE 5

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Supply current test, footnote "5/" will be renumbered as footnote "6/".

TABLE I, Single ended open loop voltage gain test, footnote "6/" will be renumbered as footnote "7/".

PAGE 6

TABLE I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Transient response test, footnote "7/" will be renumbered as footnote "8/".

TABLE I, Transient response test, TR(OS) Overshoot, for device types 05 and 06, delete "50 % max" and substitute "60 % max".

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PAGE 7

Add a new footnote "1/" which will state;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Footnote "1/" will be renumbered as footnote "2/". At the second sentence, delete the words "Device type 05" and substitute "Device types 05 and 06".

Footnote "2/" will be renumbered as footnote "3/".

Footnote "3/" will be renumbered as footnote "4/".

Footnote "4/" will be renumbered as footnote "5/".

Footnote "5/" will be renumbered as footnote "6/".

Footnote "6/" will be renumbered as footnote "7/".

Footnote "7/" will be renumbered as footnote "8/". On the first sentence, delete the words "Device type 05" and substitute "Device types 05 and 06".

PAGE 9

* FIGURE 1, add case Z as printed on pages 4 and 5 of this amendment.

* FIGURE 1, delete "Device types 01, 02, 03, 05 and 06 (cases A, C and D)* " and substitute "Device types 01, 02, 03, 05, and 06 (cases A, C, D, and Z)* ".

* FIGURE 1, delete " *14-lead flat-pack and dual-in-line (top view)" and substitute " *14-lead flat pack, dual-in-line, and gullwing flat pack (top view)".

* FIGURE 1, delete "Terminal connections." and substitute "Case outline and terminal connections."

PAGE 14

FIGURE 3, add the word "minimum" after "T_A = +125°C". Delete "+V_{IN}" and substitute "$\pm V_{IN}$".

PAGE 15

FIGURE 4, add the word "minimum" after "T_A = +125°C".

PAGE 24

Test numbers 9-16, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 9-16, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

Test number 23, adapter pin number 2, delete "5 V" and substitute "-5 V".

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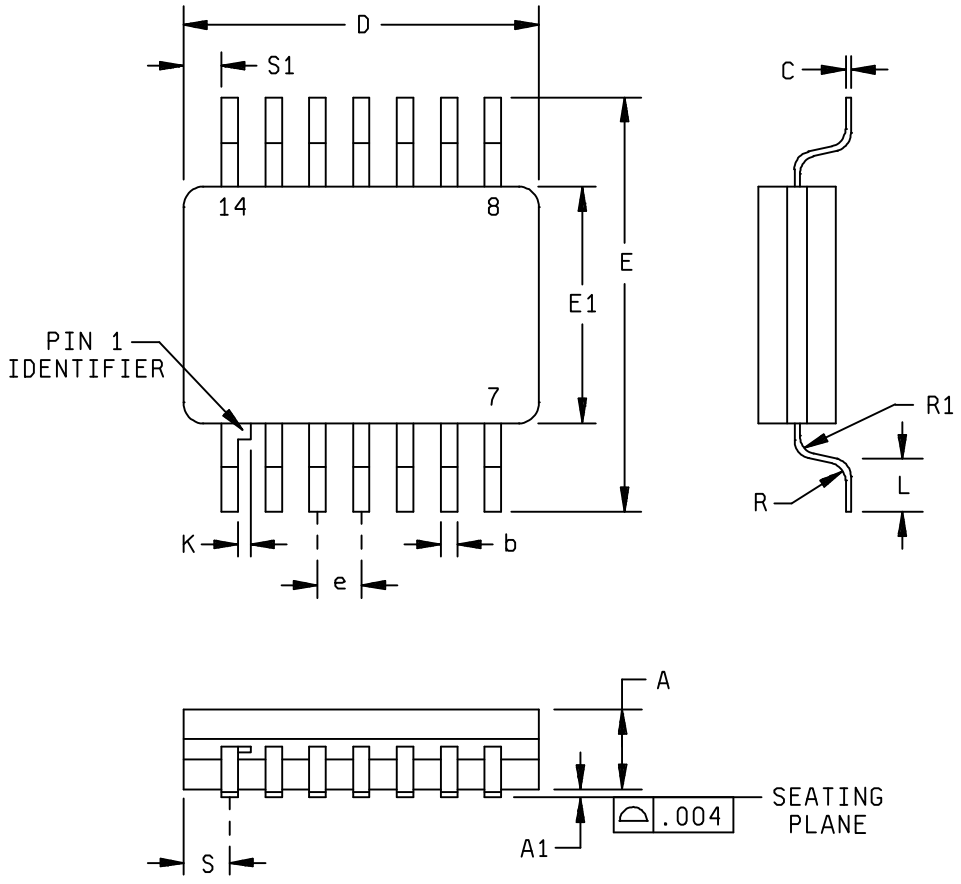
* FIGURE 1, add the following case outline:

Case Z



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* FIGURE 1, add the following case outline:

Case Z

Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.050	.080	1.27	2.03	
A1	.004	.012	0.10	0.30	
b	.015	.019	0.38	0.48	2
C	.004	.008	0.10	0.20	2
D		.390		9.91	
E		.420		10.67	
E1	.245	.270	6.22	6.86	
e	.050 BSC		1.27 BSC		
K	.007	.012	0.18	0.30	
L	.037	.043	0.94	1.09	
R	.013	.017	0.33	0.43	
R1	.013	.017	0.33	0.43	
S		.045		1.14	
S1	.005		0.13		

NOTES:
1. The US government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.
2. Maximum limit may be increased by .003 inches after lead finish is applied.

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PAGE 25

Test numbers 33-40, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 33-40, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

PAGE 26

Test numbers 57-64, device types 01 and 02, delete "+0.1 nA min" and substitute "-0.1 nA min".

Test numbers 57-64, device types 03, 04, 05, and 06, delete "-1 nA max" and substitute "+0.1 nA max".

PAGE 35

Test number 113, device types 05 and 06, delete "50 % max" and substitute "60 % max".

1/, add the following sentence; "For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

PAGE 37

4.1, delete and substitute the following;
"4.1 Sampling and inspection. Sampling inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

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TABLE IV, $+I_{IB}$ delta limits for device type 04, delete "-25 nA max" and substitute "+25 nA max".

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NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the` requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Air Force - 11, 19, 85, 99
Navy - AS, CG, MC, OS, SH

Civil Agency Coordinating Activity:
DOT-FAA(RD-650)

Preparing activity:
DLA - CC

(Project 5962-1832)

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